

Passing the Part 107 Test

Week 9 — Tuesday Lecture

The test

Questions	60 scored + 5 unscored, multiple choice A/B/C
Time	2 hours
Pass	70 % (42 of 60)
Where	PSI testing centres — locally, UVU's centre at the Provo airport
Bring	Photo ID with date of birth, signature and address
Provided	Calculator built in; the figures (on screen or the supplement book)
Retake	After 14 days, with your failed test report
Fee	*As of October 2026, about $175 — check PSI when you book*

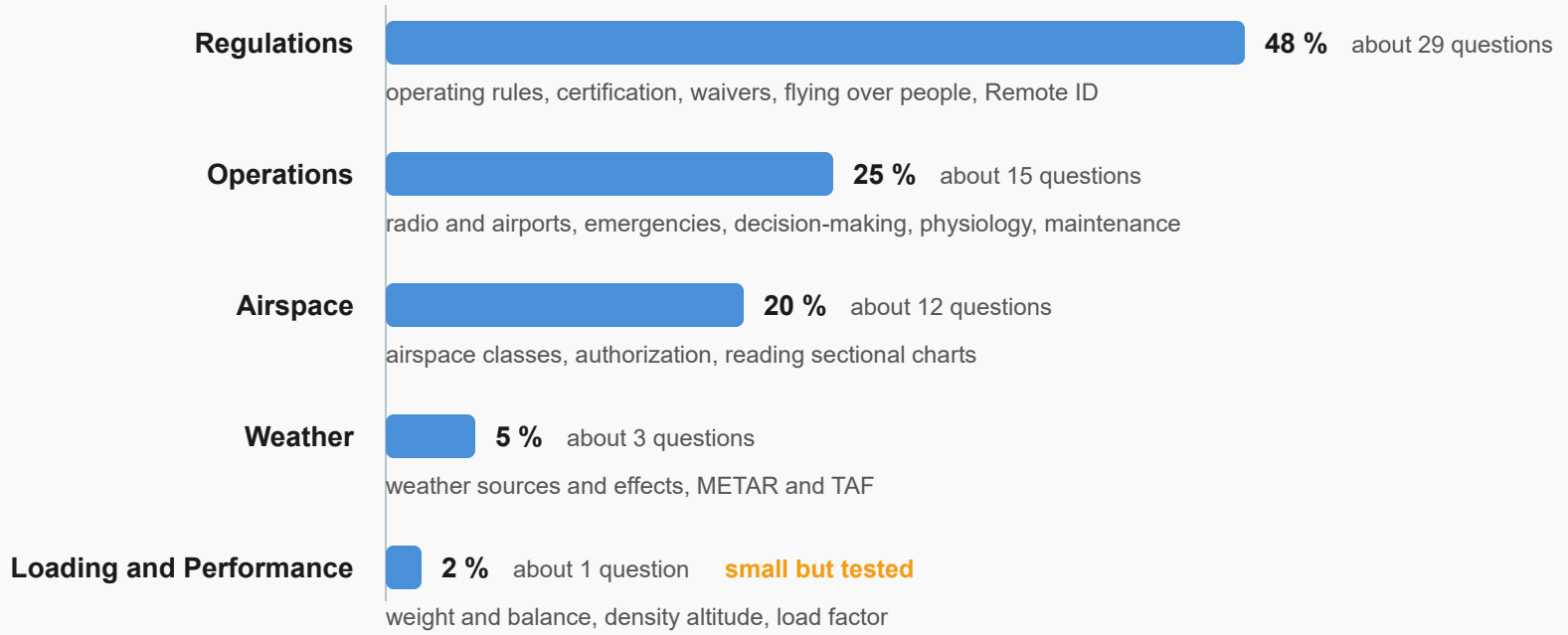
Where to start studying

#	Start with	Why	Time
1	Our readings: The Part 107 Exam and the Knowledge Review	Written for this class and current to the 2026 rules	3 h
2	AC 107-2A, chapters 5, 6 and 8	The FAA's own rules guide; regulations are half the test	3 h
3	Testing supplement chart excerpts, legend open beside them	Charts are the most-missed area; the test shows these figures	4 h
4	Matt Kendall's full video review (2024)	The one recent full review; watch one, not four	2 h
5	FAA sample questions, then timed practice exams	Real test wording; log every miss by topic	3 h

The full ranked list, a drill-by-topic table and every link: **Study Resources → Start here**. Thursday's lab walks you through finding each one.

Figure 7: What the knowledge test covers

60 scored questions. Weights from the PSI applicant bulletin, effective 29 Sept 2025; older study guides show different ranges.



Regulations alone are nearly half the test. Five unscored trial questions are mixed in, so the screen shows 65.

Half the test is regulations. Weather and loading are small, but they are on it.

How to read a figure question

"(Refer to FAA-CT-8080-2H, Figure 26, area 2.)"

1. Find **Figure 26** in the testing supplement — or on screen, inside the question
2. Find the **red number 2** on the chart
3. Read the question about *that* spot
4. When a symbol is unfamiliar, **use the legend** — it is in the supplement too



FAA-CT-8080-2H, Figure 26. Public domain. Not for navigation.

Figure 26, area 2: the red circled 2 sits beside Cooperstown airport.

Rapid round — seen in Week 3

1. Maximum altitude away from structures? A) 400 ft AGL · B) 400 ft MSL · C) 500 ft AGL
2. Minimum visibility? A) 1 SM · B) 3 SM · C) 5 SM
3. Flying 20 minutes after sunset needs? A) a waiver · B) anti-collision lights · C) nothing
4. Hours from last drink to flying? A) 4 · B) 8 · C) 12
5. Which Part 107 aircraft must be registered? A) over 0.55 lb · B) all of them · C) over 55 lb
6. Damage that must be reported? A) any · B) over $500 · C) over $1,000

ANSWER **A · B · B · B · B · B**

1. **400 ft AGL** — or 400 ft above a structure you are within 400 ft of
2. **3 statute miles** from the control station
3. **Anti-collision lights** visible 3 SM — night flying is allowed since 2021
4. **8 hours**, and below 0.04 % blood alcohol
5. **All of them** — 0.55 lb is the *recreational* threshold
6. **Over $500**, not counting the drone itself — report to the FAA within 10 days

Who issues a Declaration of Compliance?

A) The remote pilot · B) The aircraft's manufacturer, accepted by the FAA · C) The owner, after registering

ANSWER B — the manufacturer declares, the FAA accepts

- Category 2 and 3 aircraft are on an **FAA-accepted Declaration of Compliance** from the manufacturer
- The pilot's job is to **check the category on the label** before flying over people
- Category 1: 0.55 lb or less, no exposed parts that could cut skin — no declaration needed
- Category 4: an airworthiness certificate

Figure 15b: Flying over people, the four categories

§107.39 and Subpart D. No category is needed over people taking part, under a covered structure, or inside a stationary vehicle.

Category 1



0.55 lb or less

takeoff weight, everything on board

No Declaration of Compliance needed. No exposed rotating parts that could cut skin.

Weigh it with everything attached: lights, guards, payload.

§107.110

Category 2



under 11 ft·lb

impact energy on a person

FAA-accepted Declaration of Compliance from the maker, and a category label on the aircraft. No exposed parts that could cut skin.

§107.115, §107.120

Category 3



under 25 ft·lb

impact energy on a person

Declaration and label, plus: only over a closed or restricted site with people on notice, or no sustained flight over anyone not under cover.
Never over open-air assemblies.

§107.125, §107.130

Category 4



Airworthiness certificate

issued under Part 21

Certified like a manned aircraft: flown to its Flight Manual limits, maintained and inspected to its manual, records kept.

§107.140

Declaration of Compliance: the manufacturer declares it, the FAA accepts it, the label on the aircraft shows the category.

The pilot's job is to check the label. This is the most-missed item on the knowledge test (FAA statistics, 2023 to 2025).

Sustained flight over an open-air assembly (a crowd): Categories 1, 2 and 4 only with Remote ID (§89.110 or §89.115(a)); Category 3 never (§107.125).

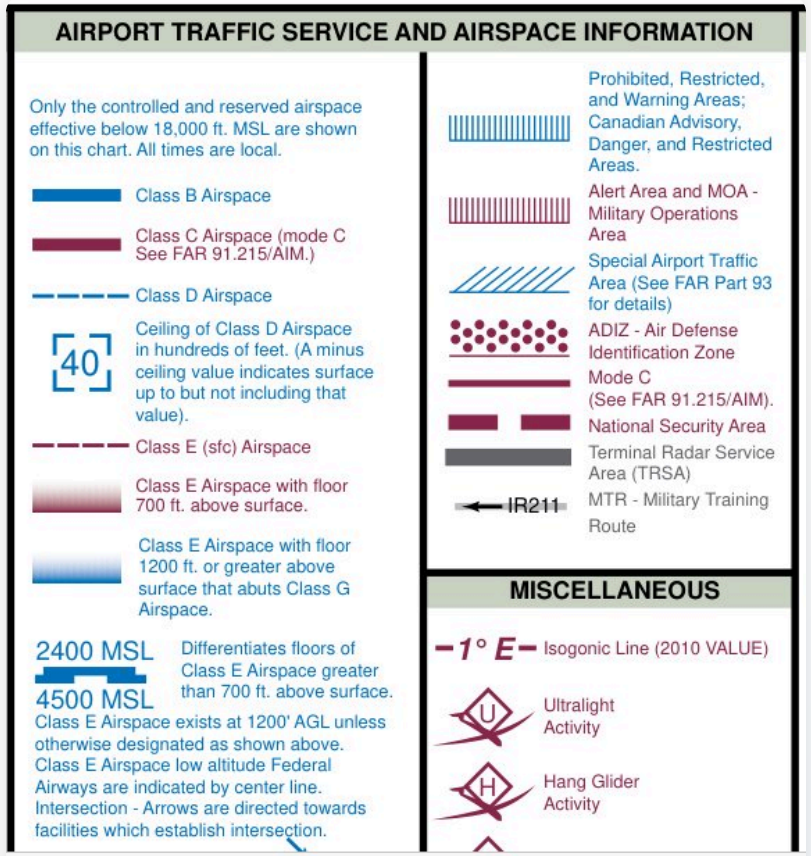
The four categories on one page. The label on the aircraft tells you which one it is.

You climb above 600 ft AGL to avoid a manned aircraft.

To whom must you report the deviation? A) Air traffic control · B) The NTSB · C) The FAA, upon request

ANSWER C — the FAA, upon request

- In an emergency you may deviate from any rule to the extent needed (§107.21)
- You report it **only if the FAA asks**
- Accidents are different: those go to the FAA **within 10 days**, whether asked or not



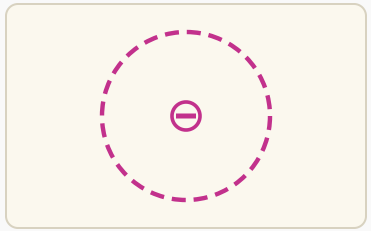
FAA-CT-8080-2H, Legend 1. Public domain. Not for navigation.

The sectional legend's airspace symbols (FAA-CT-8080-2H). When a chart symbol is unfamiliar, this is where the answer is.

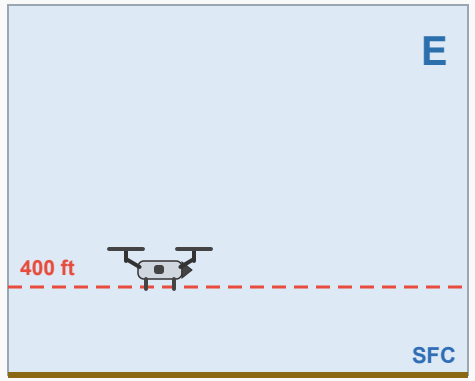
Figure 11b: Where Class E starts

Three ways the sectional draws the floor of Class E, and what each one means for a drone at or below 400 ft.

Class E from the surface



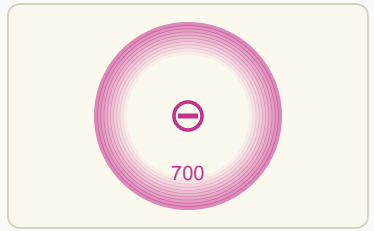
dashed magenta line



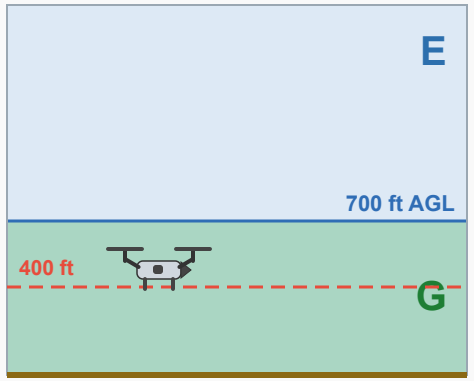
Authorization needed

Controlled from the ground up. Ask ATC first: LAANC or FAA DroneZone (14 CFR 107.41).

Class E from 700 ft AGL



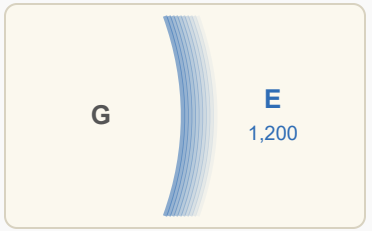
magenta vignette (shaded band)



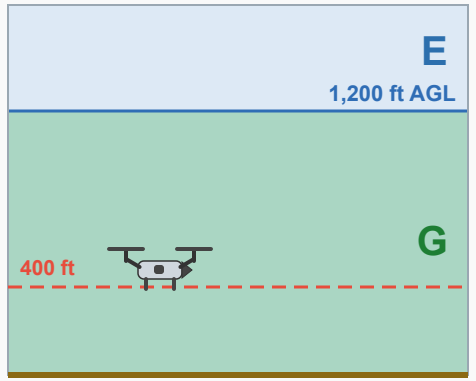
No authorization

Class G below 700 ft. Your 400 ft is uncontrolled airspace: fly, no permission needed.

Class E from 1,200 ft AGL



blue vignette, or no line at all



No authorization

The default over most of the US. Class G up to 1,200 ft, so the same answer: fly.

Only surface Class E needs authorization for a drone below 400 ft. Symbols redrawn from the chart legend, not to scale.

Class E can start at the surface, 700 ft or 1,200 ft. Only surface Class E needs authorization below 400 ft.

Savannah, area 3: what is the floor of the Class C shelf (outer circle)?

A) 1,300 ft AGL

B) 1,300 ft MSL

C) 1,700 ft MSL



FAA-CT-8080-2H, Figure 23. Public domain. Not for navigation.

ANSWER B — 1,300 ft MSL

- "41/13" means **ceiling 4,100 ft, floor 1,300 ft**, in hundreds of feet
- Airspace altitudes on a sectional are **MSL** — above sea level
- The 400 ft drone limit is **AGL** — above the ground
- **Ground elevation is on the chart:** the airport's data block ("SAVANNAH/HILTON HEAD INTL … **50** L 93": field at 50 ft), or any obstacle — "559 **(505)**" is a top at 559 ft MSL, 505 ft tall, so the ground is about 54 ft
- Shelf floor above the ground: 1,300 − 50 ≈ **1,250 ft AGL**

Inspecting a tower near Jamestown Regional, area 4. What do you need first?

A) Military authorization

B) ATC authorization

C) National Park Service authorization



FAA-CT-8080-2H, Figure 26. Public domain. Not for navigation.

ANSWER B — it is surface Class E

- The **dashed magenta line** around Jamestown marks Class E that starts at the **surface**
- Surface Class E is controlled: you need **ATC authorization** (§107.41), usually through LAANC
- A **magenta** shaded edge means Class E from 700 ft; **blue** means 1,200 ft — no authorization needed below 400 ft

Area 2: the gray line labelled "VR1667, VR1617…". Is it a hazard?

A) No, all operations are above 400 ft

B) Yes, a military training route from the surface to 1,500 ft AGL

C) Yes, the route separates traffic



FAA-CT-8080-2H, Figure 59. Public domain. Not for navigation.

ANSWER B — a military training route, surface to 1,500 ft AGL

- **VR** = flown under visual flight rules; **IR** = instrument rules
- **Four digits**: at or below 1,500 ft AGL. **Three digits**: some segments above 1,500 ft
- Military jets fly these fast and low — be alert near them, though they are not off-limits

So can you fly a survey under that route at 300 ft?

It is a hazard. Is it allowed? What would you do first?

ANSWER Yes — legal, but plan for the jets

- An MTR is **not restricted airspace**: no permission is needed beyond the airspace class underneath
- The jets can fly **as low as 100 ft AGL** at **well over 250 knots** — your 400 ft ceiling overlaps their route
- The line is a **centreline**; the route can be several miles wide
- You must **yield** (§107.37), and a jet can close a mile in about ten seconds
- So: ask Flight Service (1-800-WX-BRIEF) if the route is active, add a visual observer, stay low, and be ready to land

A crop survey in the Devils Lake West MOA. How do you find out whether it is active?

A) The chart legend

B) A small UAS database

C) A military operations directory



FAA-CT-8080-2H, Figure 21. Public domain. Not for navigation.

ANSWER A — the chart legend

- A full sectional lists each special-use area's times of use in its legend and margin tables
- A **military operations area (MOA)** is a hazard to know about, not a ban
- A **restricted area** needs the controlling agency's permission
- A **TFR — temporary flight restriction** — closes airspace for a while: a wildfire, a stadium game, a VIP visit. It is **not printed on the chart**
- A **NOTAM — notice to air missions** — is the FAA's bulletin of short-term changes, and it is how TFRs are announced
- Check both before every flight: tfr.faa.gov, or your LAANC app

Area 2: what is the CTAF at Coeur d'Alene?

A) 122.05

B) 135.075

C) 122.8



FAA-CT-8080-2H, Figure 22. Public domain. Not for navigation.

ANSWER **C — 122.8**

- **CTAF**: the common traffic advisory frequency, where pilots announce their positions at an airport without an operating tower
- On the chart it is the frequency with a **C in a circle** after it
- A drone pilot doesn't have to transmit, but **listening** tells you where manned traffic is

Figure 11c: Reading a traffic pattern

Runway 13, standard left-hand pattern, north up. Where is an aircraft that reports “midfield left downwind, runway 13”?

How to read it

Runway number = heading ÷ 10

Runway 13 lands toward 130° magnetic. Runway 31 is the same strip from the other end, heading 310°.

Standard pattern: all turns left

Unless the chart says RP, right pattern.

Downwind: parallel, opposite way

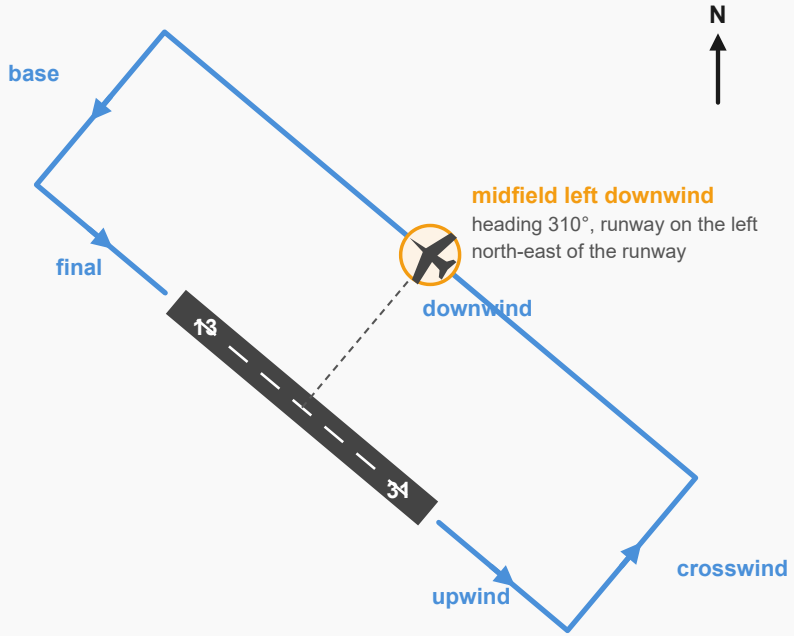
The runway stays on the pilot’s left, so the downwind of runway 13 is flown on 310° along the north-east side.

The test answer

“Midfield left downwind, runway 13” puts the aircraft east of the runway (FAA sample question, Figure 26, area 2).

Why a drone pilot cares

Stay out of the pattern (14 CFR 107.43). Hear a position report, know where to look.



Upwind, crosswind, downwind, base, final: five legs, four left turns. Draw the pattern from the runway number and the answer falls out.

Runway 13 points about 130° magnetic. A left pattern turns left; downwind runs parallel, the opposite way.

Midfield left downwind — the test version

(Refer to FAA-CT-8080-2H, Figure 26, area 2.) While monitoring the Cooperstown CTAF you hear an aircraft announce that they are midfield left downwind to RWY 13. Where would the aircraft be relative to the runway?

A) The aircraft is East.

B) The aircraft is South.

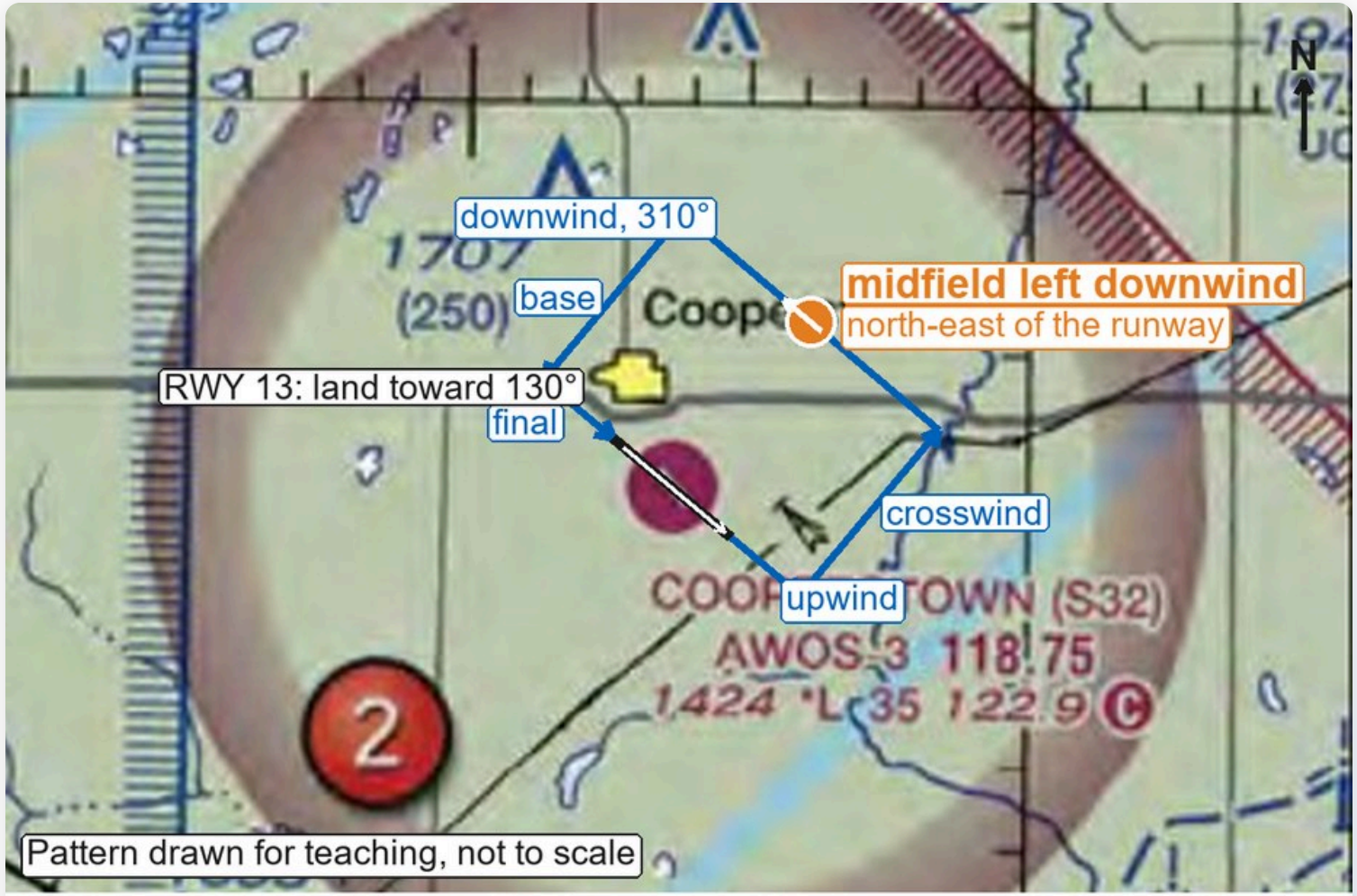
C) The aircraft is West.



FAA-CT-8080-2H, Figure 26. Public domain. Not for navigation.

ANSWER A — east of the runway

- Runway 13: landing toward about **130°** magnetic
- Left traffic: every turn is to the **left**, so the pattern sits on the **left** of the landing direction
- Downwind runs the opposite way, **310°**, on that side — **north-east**, so "east" of the runway
- On the test: find the runway, draw the landing arrow, turn left four times



FAA-CT-8080-2H, Figure 26. Public domain. Not for navigation.

Figure 24: Five hazardous attitudes and their antidotes

Recognise the thought, name the attitude, say the antidote. Wording from the FAA Pilot's Handbook, chapter 2.

The attitude, and what it sounds like	The antidote: say it to yourself
1 **Anti-authority** *"Don't tell me."*	**Follow the rules. They are usually right.**
2 **Impulsivity** *"Do it quickly."*	**Not so fast. Think first.**
3 **Invulnerability** *"It won't happen to me."*	**It could happen to me.**
4 **Macho** *"I can do it."*	**Taking chances is foolish.**
5 **Resignation** *"What's the use?"*	**I'm not helpless. I can make a difference.**

The test tells a short story and asks which attitude it shows. "Fly first, ask questions later" is impulsivity; "watch this" is macho; "the rules are for other people" is anti-authority.

Five hazardous attitudes, and what to tell yourself instead.

Which hazardous attitude?

You have been hired as a remote pilot by a local TV news station to film breaking news with a small UA. You expressed a safety concern and the station manager has instructed you to "fly first, ask questions later." What type of hazardous attitude does this attitude represent?

A) Machismo. · B) Invulnerability. · C) Impulsivity.

ANSWER C — impulsivity

- Impulsivity: "Do it quickly." Antidote: **"Not so fast. Think first."**
- Machismo (macho) is "I can do it"; invulnerability is "It won't happen to me"
- On the test, match the **words** to the attitude

Figure 23 — IMSAFE Personal Fitness Checklist

I - M - S - A - F - E

I	**Illness**	Am I suffering from any illness or symptom of illness? Do not fly if sick.
M	**Medication**	Am I taking any medications that could impair judgment or reaction time?
S	**Stress**	Am I under psychological pressure? Stress narrows attention and impairs decisions.
A	**Alcohol**	8 hours bottle-to-throttle. No flying under the influence (0.04% BAC limit per FAR).
F	**Fatigue**	Am I adequately rested? Fatigue degrades performance as much as alcohol.
E	**Emotion / Eating**	Am I emotionally upset or hungry/dehydrated? Strong emotions impair judgment.

If the answer to ANY of these is "yes," consider postponing the flight.

IMSAFE: check yourself before you check the aircraft.

A battery is swollen and warm before a flight.

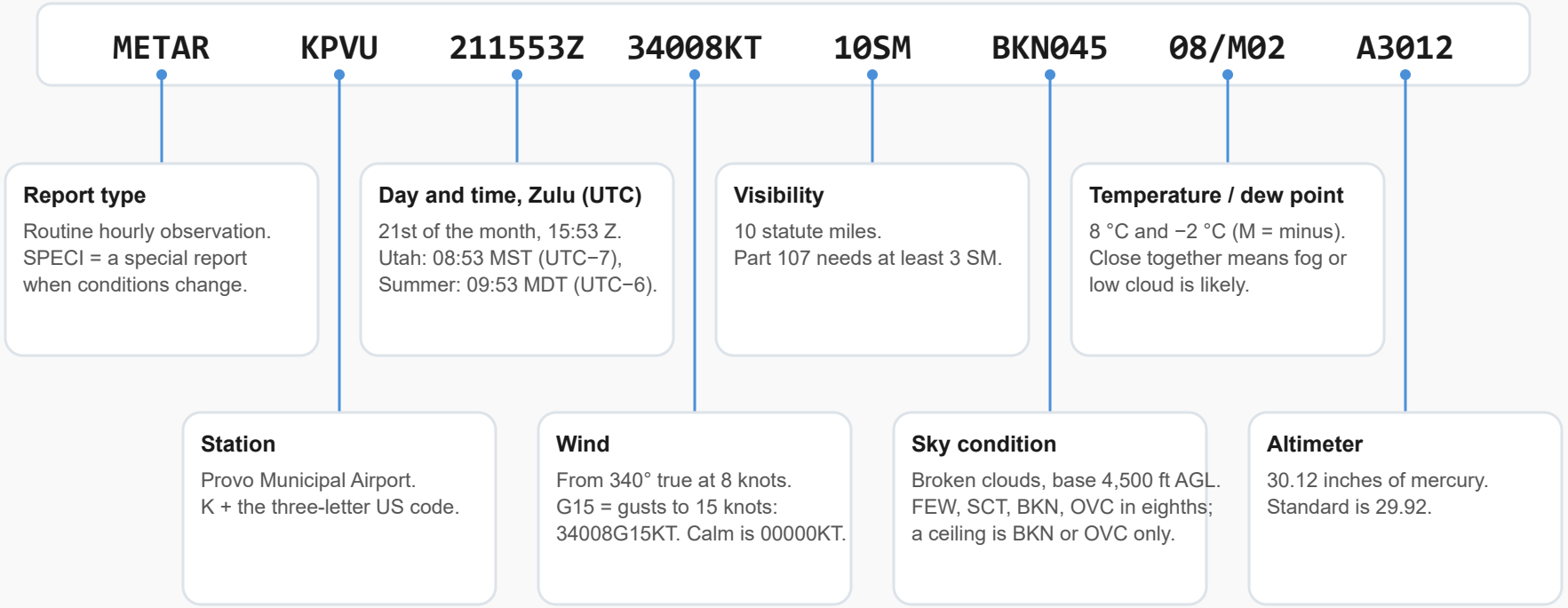
What do you do? A) Fly a short flight · B) Cool it in the fridge · C) Don't fly it; move it outdoors onto a non-flammable surface

ANSWER C — never fly a damaged lithium battery

- Swelling, heat or smell means the cell is damaged and can catch fire
- Move it **outdoors**, onto sand or concrete, away from people and anything that burns
- Follow the manufacturer's disposal instructions
- Lithium fires are hard to put out — keep a fire bag or sand bucket in the kit

Figure 16: Reading a METAR

An example in the real format, not an actual observation: a winter morning at Provo. Read it group by group.

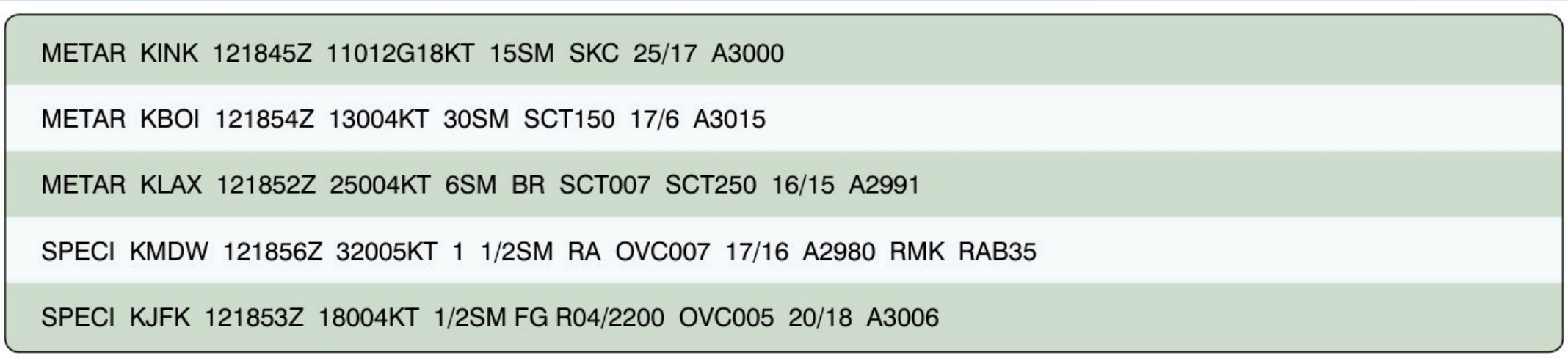


A METAR, decoded. Weather is about 5 % of the test.

Current conditions at Chicago Midway (KMDW)?

(Refer to FAA-CT-8080-2H, Figure 12.) The full report is the **SPECI KMDW** line in the table — find it first.

A) Sky 700 feet overcast, visibility 1-1/2SM, rain. · B) Sky 7,000 feet overcast, visibility 1-1/2SM, heavy rain. · C) Sky 700 feet overcast, visibility 11, occasionally 2SM, with rain.



METAR KINK 121845Z 11012G18KT 15SM SKC 25/17 A3000
METAR KBOI 121854Z 13004KT 30SM SCT150 17/6 A3015
METAR KLAX 121852Z 25004KT 6SM BR SCT007 SCT250 16/15 A2991
SPECI KMDW 121856Z 32005KT 1 1/2SM RA OVC007 17/16 A2980 RMK RAB35
SPECI KJFK 121853Z 18004KT 1/2SM FG R04/2200 OVC005 20/18 A3006

FAA-CT-8080-2H, Figure 12. Public domain. Not for navigation.

ANSWER A — overcast at 700 ft AGL, 1½ statute miles, rain

- Cloud height in **hundreds of feet AGL**: 007 = 700 ft
- A **ceiling** is the lowest **broken (BKN)** or **overcast (OVC)** layer
- Visibility below 3 SM: **no Part 107 flight**
- **RA** alone is moderate rain; **+RA** would be heavy
- Watch for: times are **Zulu** (Utah is UTC−7 in winter, −6 in summer); winds are **true**

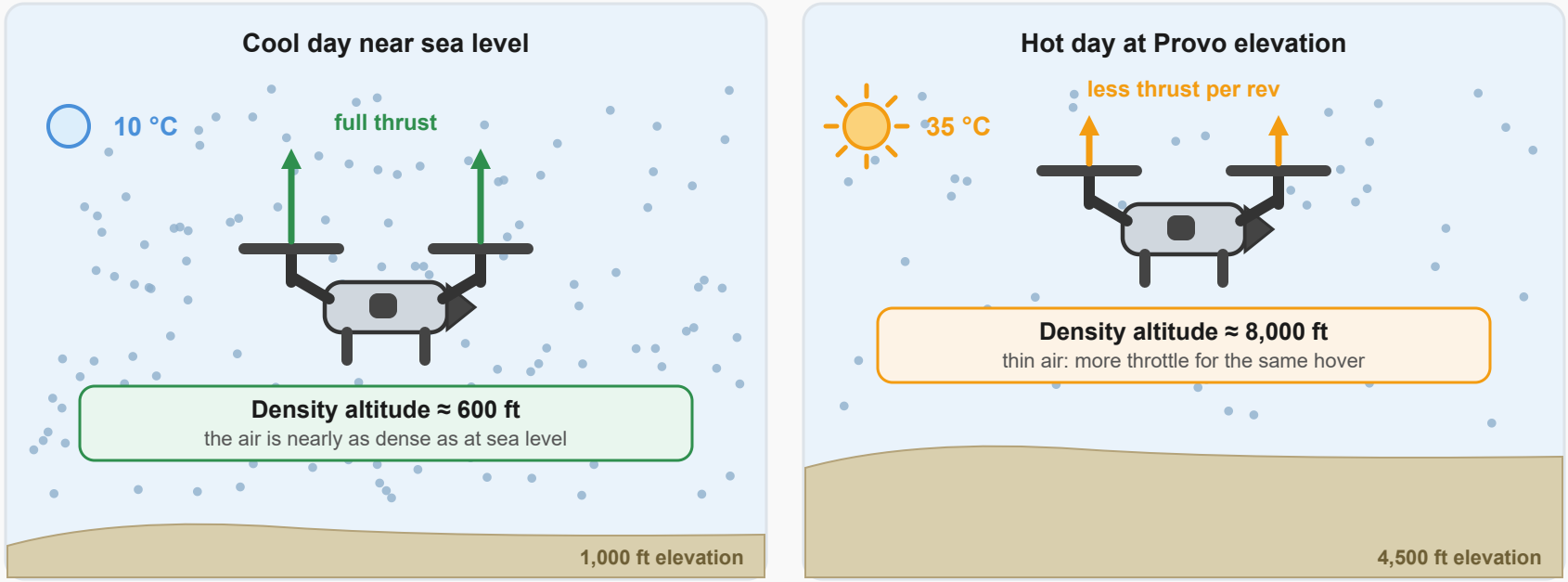
Decoding the KMDW report, group by group

```
SPECI KMDW 121856Z 32005KT 1 1/2SM RA OVC007 17/16 A2980 RMK RAB35
```

Group	Means
`SPECI`	A special report, issued because conditions changed (a routine one says METAR)
`KMDW`	Chicago Midway International Airport
`121856Z`	12th of the month, 18:56 Zulu (UTC)
`32005KT`	Wind from 320° true at 5 knots
`1 1/2SM`	Visibility 1½ statute miles — below the 3 SM Part 107 needs
`RA`	Rain, moderate (`-RA` light, `+RA` heavy)
`OVC007`	Overcast at 700 ft above the ground — this is the ceiling
`17/16`	Temperature 17 °C, dew point 16 °C — only 1° apart, so fog or low cloud is likely
`A2980`	Altimeter setting 29.80 inches of mercury
`RMK RAB35`	Remarks: rain began at 35 minutes past the hour

Figure 19: Hot and high: density altitude

The same aircraft on two days. Thin air gives the propellers less to push on, and the aircraft flies as if it were much higher than it is.



High density altitude reduces propeller efficiency and performance.

Less thrust and less lift: a weaker climb, more throttle to hover, a shorter flight, and a longer takeoff for a fixed wing.

Density altitude rises with elevation, temperature and humidity. Rule of thumb: about 120 ft per °C above the standard day (15 °C at sea level, 2 °C cooler per 1,000 ft). Pilot's Handbook of Aeronautical Knowledge, chapter 11.

Hot and high: thinner air means less lift. Loading and performance is about 2 % of the test.

A 33 lb unmanned airplane in a 30° level banked turn.

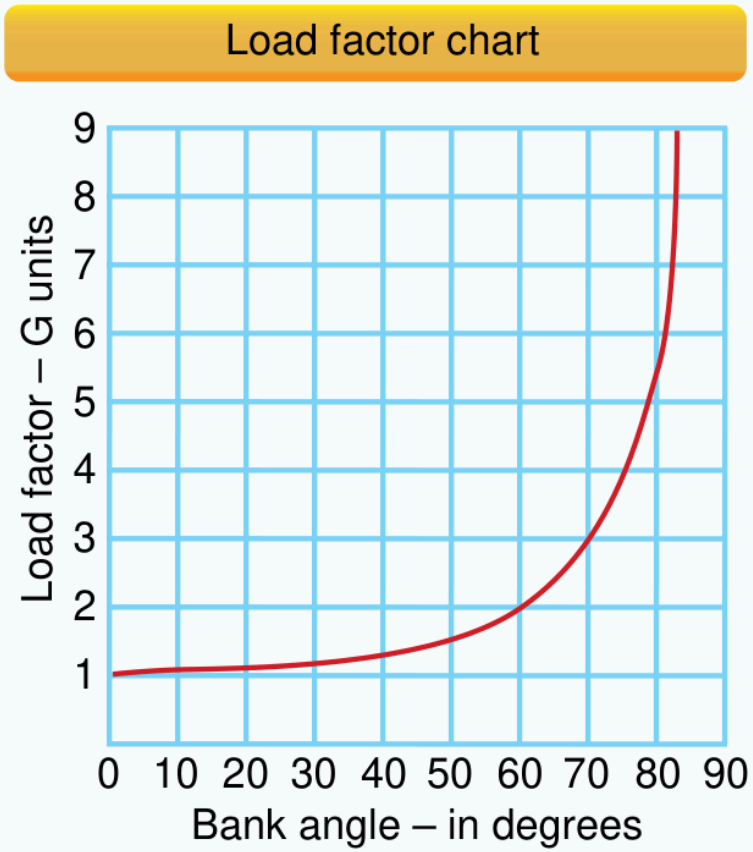
What load must its structure support?

A) 34 lb

B) 47 lb

C) 38 lb

Angle of bank ϕ	Load factor n
0°	1.0
10°	1.015
30°	1.154
45°	1.414
60°	2.000
70°	2.923
80°	5.747
85°	11.473
90°	∞

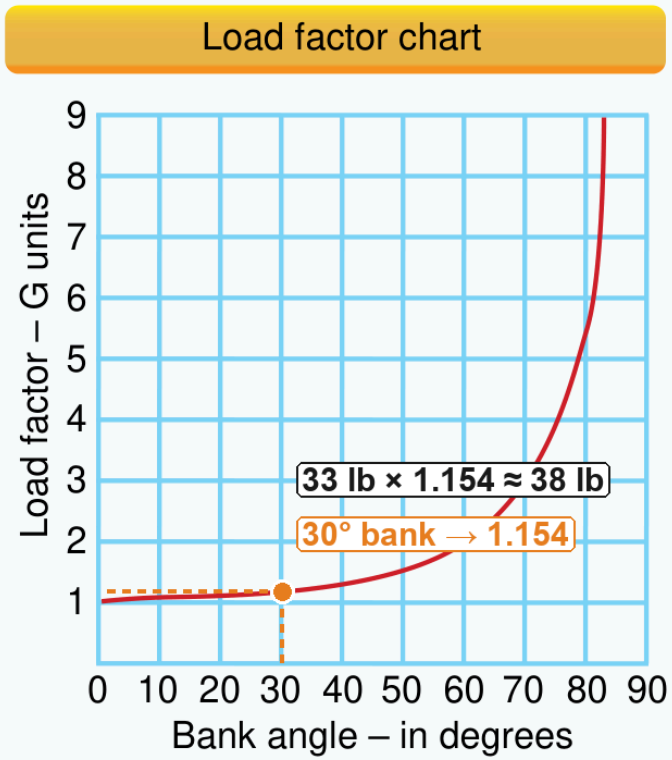


FAA-CT-8080-2H, Figure 2. Public domain. Not for navigation.

ANSWER C — about 38 lb

- Find **30°** in the table, or go up from 30° to the curve: load factor **1.154**
- 33 lb × 1.154 ≈ **38 lb**
- 45° → 1.41; 60° → 2.0 — steep turns get expensive fast
- The table is the faster read; the curve shows why steep turns are risky

Angle of bank φ	Load factor n
0°	1.0
10°	1.015
30°	1.154
45°	1.414
60°	2.000
70°	2.923
80°	5.747
85°	11.473
90°	∞



FAA-CT-8080-2H, Figure 2. Public domain. Not for navigation.

The traps

- Chart altitudes are **MSL**; cloud heights and the 400 ft limit are **AGL**
- A **ceiling** is BKN or OVC only
- METAR times are **Zulu**; METAR winds are **true**
- **Surface** Class E needs authorization; 700 ft and 1,200 ft Class E don't, below 400 ft
- MOA = hazard; restricted = permission; temporary flight restriction (TFR) = announced by NOTAM, not on the chart
- Emergency deviations: report to the FAA **on request**; accidents: **within 10 days**

A two-week study plan

- **Week 1:** the overview reading and the knowledge review; FAA sample questions; AC 107-2A
- **Week 2:** charts every day — the supplement figures and the legend; the practice exams here
- **Day before:** the traps list; sleep
- Free: FAA sample questions · AC 107-2A · the Pilot's Handbook of Aeronautical Knowledge · PSI's sample test
- Knowledge Review · Study Resources — Start here · ranked list on slide 3

What to do Thursday

- **Part 107 study lab:** Part 107 Studying
- The **practice exam** opens this week — see Learning Suite
- Practice: Mini Practice Exam · Full Practice Exam